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JavaScript - TEST

Batch: 8 to 10

1. Write a program to print all even numbers from 1 to 50 using a for loop in single line.

```
for ( let i=1; i<=50; i++ ){
    if (i%2 === 0){
        Console.log(i);
    }
}
```

2. Write a program to find the factorial to given number using a while loop.

```
let num = 5;
let factorial = 1;
while (num >0) {
    factorial *= num;
    num--;
}
Console.log (factorial);
```

Output:-

120

3. Write a program to print numbers from 20 to 1 using do-while loop.

```
let i= 20;
do {
    Console.log(i)
```

```
    i--;
} while (i>=1);
```

(4) Write program to reverse the string "JavaScript" using the built in reverse () method.

```
let language = "JavaScript";
let reversed = " ";
for ( let i= language.length-1; i>=0; i--){
   reversed += language[i];
}
   console.log (reversed);
```

Output :-

tpircSavaJ

(5) Give the following array, find the highest number using a loop.

```
const numbers = [45, 12, 89, 34, 67, 90, 23];

let highest = numbers[0];
for (let i=1 ; i< numbers.length; i++) {
   if (numbers [i] > highest){
      highest = numbers [i];
   }
}
   console.log (highest);
```

Output:-

90.

(6.) Give the following array. print only the odd numbers and find their total.

```
Const numbers=[10,15,20,25,30,35,40];

let total = 0;

for (let i=0; i<numbers.length; i++){
if (numbers[i]%2 != =0){
  console.log(numbers[i]);
   total += numbers[i];
  }
}

   Console.log("Total", total);
```

(7) Create an Object called Student with name, age, course, and mark. Access and display all values.

```
Const student = {
            name: "Tabasiya",
            age: 22,
            course: "JavaScript",
            mark: 92
     };
       Console.log(Student.name);
       Console.log(student.age);
       Console.log(student.course);
       Console.log(Student.mark);
```

⑧ Given the following array of object, display only the students who scored more than 75.

```
const Students=[
    {name:"Tabas", mark: 85 },
    name: " Sameera", mark: 65 },
    name:"Chasan", mark: 90},
    name: "Deepak" mark: 70}
];
for (let Student of students){
  if (Student.mark > 75){
    Console.log (Student.name, Student.mark);
  }
}
```

⑨ Create three different functions to calculate the square of a number: i.) Normal function ii.) Function Expression iii.) Arrow function.

i.) Normal function:-

```
function display(name){
    return " Hello "+name;
}
Console.log (display (" Tabas"));
```

Output:-

Hello Tabas.

ii.) Function Expression:-

```
const add = function (a, b){
    return a+b;
};
Console.log (add (10,20));
```

Output:-

30

(iii) Arrow function:-

```
const square = (num) => {
return num * num;
};
console.log (square(5));
```

output :-

25.

(10) Create a function that accept name and age as parameters and return a formatted sentence using a template literal.

Expected format:

My name is Tabasiya and I am 25 years old.

```
function display (name, age) {
   return `my name is ${name} and I am ${age}
years old.`;
}
   console.log (display ("Tabasiya", 25));
```

Output:-

My name is Tabasiya and I am 25 years old.

(11) Write a program demonstrating the difference b/w var, let, and const inside a block scope.

```
var \ =10;
{
    var a = 20;
    let b= 30;
    const c= 40;
    console.log (a);
    console.log (b);
    console.log (c);
}
console.log(a);  → var is accessible
                   outside the block.
```

(12) Predict the output and explain the reason.

```
console.log (a);
var a=10;
```

Also, explain what happens var is replaced with let.

```
console.log(a);
var a= 10;
```

outputs-

undefined

→ When var is replaced with let,

```
console.log (a);
let a= 10;
```

Out put :-

Reference Error.

(13) Given the following Object, extract name and Course using destructing.

```
Const student = {
    name: "Ravi",
    age : 22
    Course: "JavaScript"
};

Const {name, Course } = Student;
Console.log (name);
Console.log (Course);
```

Output:-
Ravi
JavaScript

(14.) Create an arrow function using a rest parameter that accepts multiple numbers their ~~sum~~ return their total.

Example:-
Input: 10, 20 , 30, 40
output : 100

```
Const total= (...numbers) => {
    return numbers.reduce ((Sum, num)
    => Sum +num, 0);
};
Console.log (total (10,20,30,40));
```

Output:-
100

(15) Given the following data:-

```
const users = [
    {name: "Ravi",
     marks: [80, 90, 85],
     address: {
         city: "Chennai"
     }
    },
    {
        name: "Kumar",
        mark: [60, 70, 75]
    }
];
```

Write program to display each user's name, city, and total marks. If the city does not exist, display:

City Not Available.

use modern JS features wherever possible.

Ans:

```
for (const user of users){
    const {name, marks, address} = user;
    const city = address?.city ??
    "City Not available";
    const total = marks.reduce((sum, mark)
    => sum + mark, 0);
    console.log(name, city, total);
}
```

output:-

Ravi Chennai 255.

Kumar city not available 205.